

Muhammad **Mueed-Ul-Haq**

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Summary

Power engineering graduate with 6+ years of experience in power electronics controls, grid simulations, and energy storage systems. Currently, working as a Senior Power Electronics Controls Engineer, leading technical development and working closely with cross-functional teams in fast-paced R&D environments. Skilled in analyzing distribution networks and using data to improve energy reliability and support smarter grid planning.

Work Experience

Fluence (A Siemens and AES Copmany), Power Conversion Systems Team
Erlangen, Germany

Senior Power Electronics Controls Engineer
Apr. 2025 - Present

- Driving development of innovative solution around grid-forming application controls for PCS in utility scale energy storage systems.
- Designing and developing cutting-edge control algorithms for power electronic systems used in utility-scale energy storage systems, such as inverters and DC/DC converters.
- Utilizing various simulation tools to model, simulate and analyze power electronics systems under various operating conditions.
- Implementing and validating control strategies through prototyping, testing, and iterative optimization to ensure optimal system performance.
- Execute hardware-in-the-loop (HIL) and software-in-the-loop (SIL) testing to validate control algorithms, ensuring compatibility with complex grid scenarios.
- Responsibilities also include to act as the technical lead in R&D programs, to provide mentorship to junior controls engineers.

E.ON Group Innovation, R&D Team
Munich, Germany

Power Electronics Engineer
Dec. 2023 - Mar. 2025

- Lead role for the design and implementation of advanced Battery Energy Storage Systems (BESS).
- Develop and optimize grid performance strategies and innovative power electronics solutions.
- Successfully identified two research challenges and technological gaps in the field of BESS and power electronic inverters for grid integration.
- Filed patents for the developed technologies, protecting intellectual property and advancing state-of-the-art solutions.
- This position also required expertise in energy markets, utility grid interconnection standards and regional grid codes to navigate industry requirements effectively.

E.ON Digital Technology, R&D Team
Munich, Germany

Working Student
Nov. 2022 - Nov. 2023

- Performed detailed simulations of IEEE test feeders using OpenDSS, PowerPanda, and MATLAB, covering a range of fault conditions, load variations, and grid configurations.
- Generated simulation data sets under diverse operating scenarios to support system behavior analysis and model validation.
- Analyzed system response to faults and load changes to identify performance issues and assess reliability metrics.
- Cross-validated results across different simulation platforms to ensure accuracy and consistency in grid modeling.

TDK Electronics, Magnetics Department
Munich, Germany

Working Student
May. 2022 - Oct. 2022

- Designed and analyzed magnetic components for complex electrical systems as part of the magnetics team.
- Designed and tested various types of common mode chokes.
- Obtained and analyzed lab measurements for product datasheets.

Power Technology Research, E-Mobility Team
Lahore, Pakistan

Analyst
Aug. 2020 - Apr. 2022

- Conducted market research on global electric vehicles for Fortune 500 companies.
- Published articles and infographics on e-mobility.
- Presented research at the Enlit Europe conference, Milan, 2021.

BarqBox
Lahore, Pakistan

Research Associate
May. 2020 - Aug. 2020

- Worked on the development of IoT enabled Li-Ion batteries for light commercial vehicles.
- Implemented remote switching and monitoring functionality for the batteries.

Education

Technical University of Munich — *Munich, Germany*

M.Sc in Power Engineering — *Oct. 2021 - Nov. 2023*

- Grade: 1.8/1
- Relevant Courses: Power Electronics, Advanced Controls, Energy Storage, Renewable Energy Technology, Integration of Renewable Energies

Lahore Unviersity of Managmenet Sciences (LUMS) — *Lahore, Pakistan*

BS Electrical Engineering — *Aug. 2016 - May. 2020*

- CGPA: 3.63/4.0, German Grade: 1.37/1
- Relevant Courses: Power Electronics, Advanced Controls, Energy Storage, Renewable Energy Technology, Integration of Renewable Energies

Projects

Development of Advanced Charging and Discharging Strategies for Battery Energy Storage Systems to Mitigate Degradation — *Munich, Germany*

E.ON SE & the Chair of Electrical Energy Storage at the Technical University of Munich — *Apr. 2024 - Mar. 2025*

- Led a project team focused on developing intelligent solutions to improve battery availability on the grid while minimizing degradation.
- Explored autonomous bidding strategies for market participation to help battery owners maximize profits.
- Applied AI techniques for predictive maintenance to reduce the risk of battery failures and prevent grid disconnections.

3-Phase Adaptive Inverter Controller Design using ML-based techniques — *Munich, Germany*

E.ON SE & the Chair of High Power Converters at the Technical University of Munich — *May. 2023 - Nov. 2023*

- Designed an adaptive power electronic inverter using state-of-the-art grid forming and grid following controllers and AI and Machine Learning to improve grid stability and performance.
- Replaced conventional inverter control techniques to enhance grid dynamics.
- Facilitated easier integration of renewable energy into the existing grid.

Active Balancing of Li-Ion Cells Using Analog Circuit Design — *Lahore, Pakistan*

Lahore University of Management Sciences (LUMS) — *May. 2019 - May. 2020*

- Implemented a scalable Battery Management System (BMS) using imbalance-driven current to balance cells at variable rates.
- Designed, built, and tested a bi-directional Cuk converter for energy transfer between cells.
- Developed a cost-effective analog control circuit for the BMS as an alternative to existing solutions.

Honors & Awards

Academic Honors

2020 **Award of Distinction,** Graduated with a Distinction in Electrical Engineering from LUMS — *Lahore, Pakistan*

2020 **Dean's Honor List,** Received a Dean's Honor List Award from LUMS — *Lahore, Pakistan*

Presentation

Enlit Europe 2021 — *Milan, Italy*

Presenter on the topic of Substation Automation — *Dec. 2021*

- Presented trends in the centralized vs. distributed utility architecture.

Writing

Detachable Lithium-Ion Starter Battery and Analog Battery Management System for ICE Vehicles — *2021 IEEE Fourth International Conference on DC Microgrids (ICDCM)*

First Author — *Aug. 2021*

- Presented my research on a novel design methodology for a scalable BMS that employs active balancing using analog discrete components.
- The analog control system of a BMS integrated with a DC-DC bi-directional Cuk converter allows increased reliability and efficiency.